PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Blood Bank Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Harsimran Kaur	02-10-2026	Initial draft: prepared all sections of the SRS (Introduction, Overall Description, External Interface Requirements, Analysis Models, Non-Functional Requirements, Appendices) and wrote the functional requirements .	1.0
Chukkapalli Rohan	02-10-2026	Added functional requirements for Donor Management (5.3) and Search Blood and Donors (5.4)	1.1

Introduction

Purpose

This document specifies the software requirements for the Blood Bank Management System (BBMS), version 1.0. It covers the complete system from user accounts, donor management, blood donation recording, blood stock management, blood request handling, search and reports. It is written to the IEEE SRS structure and is the basis for the architecture and design specification, the test plan and the implementation.

Intended Audience

The SRS is intended for the following readers:

1) *Developers, who use it to build features and check if all the requirements are met.*
2) *Testers, who derive the test cases from the requirements.*
3) *Faculty and reviewers who evaluate the project*
4) *Project team members who use it to keep the scope and ownership clear.*

Section1 deals with the introduction to the product. Section2 describes the product and the constraints. Section 3 describes the interfaces. Section 4 gives the use case model. Section 5 lists the functional requirements .Section 6 lists the non functional requirements . Section 7 and the appendices contain the glossary, field layout and traceability matrix.

Product Scope

The BBMS is a web-based system that helps a blood bank manage donors, donations, blood stock and blood requests from hospitals and patients. It replaces paper registers and spreadsheets.

Objectives:

- *Keep an accurate, up-to-date record of donors and of blood units by blood group and expiry date.*
- *Let hospitals and patients find available blood and request it online.*
- *Reduce wastage by warning staff about low stock and units close to expiry.*
- *Protect donor and patient data through role-based access.*

Out of scope for version 1.0: online payments, SMS or push notifications, live location tracking, blood component separation, blood group compatibility substitution and a mobile application. Some of these are listed as future enhancements in Section 7.

References

- *IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications, and ISO/IEC/IEEE 29148:2018.*

- *PES University, Software Engineering (UE24CS341A) Mini-Project Problem Statement, Section C, Team 9: Blood Bank Management System.*
- *Software Engineering Mini-Project Guidelines and Deliverables Part-1, PES University, 2026.*
- *Digital Personal Data Protection Act, 2023 (India).*
- *Applicable national blood donor selection and blood storage guidelines.*

Overall Description

Product Perspective

The BBMS is a new, self-contained web application. It does not replace or connect to an existing software system. It has four parts that work together: a browser-based user interface, an application server that holds the business rules, a relational database that stores all records, and an email service used for password reset links.

Product Functions

At a high level the system lets its users:

- *Register, log in and manage their own profile (5.1).*
- *Manage user accounts and roles as an administrator (5.2).*
- *Maintain donor records and check donor eligibility (5.3).*
- *Search stock and donors by blood group and location (5.4).*
- *Record blood donations and create blood units (5.5).*
- *Track blood stock, expiry and low-stock alerts (5.6).*
- *Submit, review, approve, reject and issue blood requests (5.7).*
- *Generate and export reports (5.8).*

User Classes and Characteristics

User class	***Description and characteristics***	***Main functions used***
Donor	*Member of the public who donates blood. Occasional use, no technical training, mostly on a phone.*	*Register/login, profile, own donation history*
Hospital / Patient	*Hospital staff or patient family who need blood. Occasional, often urgent use. Basic computer skills.*	*Register/login, search stock, submit and track requests*

Blood Bank Staff	*Employees who run the blood bank. Daily use, trained on the system. The most important user class.*	*Donor records, donation recording, stock, request processing, reports, search*
Admin	*Manages accounts and system settings. Infrequent use, technically skilled.*	*User and role management, audit log, reports, configuration*

Operating Environment

- *Client: a web browser (the latest two versions of Chrome, Firefox or Edge) on Windows, Linux, macOS or Android.*
- *Server: any Linux or Windows machine able to run the chosen backend (Node.js or Python/Django; to be fixed in the architecture document).*
- *Database: a relational database such as PostgreSQL or MySQL.*
- *Network: HTTPS access over the Internet or a local network.*

Design and Implementation Constraints

- *The system shall be implemented in one of the languages allowed by the problem statement: JavaScript (Node.js) or Python (Django).*
- *The system shall store donor and patient data in a relational database and follow the security requirements in Section 6.*
- *Donor eligibility limits and blood shelf life shall be configurable rather than fixed in the code.*
- *The project must be completed within the semester schedule by a team of four students.*

2.6 Assumptions and Dependencies

- *Users have a device with a modern browser and an Internet connection.*
- *Donor eligibility limits (age 18 to 65, weight 45 kg, 90/120-day gap) and a 35-day shelf life are used as defaults and will be confirmed against the applicable guidelines.*
- *Each hospital or patient request is for one blood group; automatic substitution by compatibility is not provided in version 1.0.*
- *An email service is available for password reset links.*
- *Hospital and patient identity is verified outside the system by blood bank staff before blood is handed over.*

External Interface Requirements

User Interfaces

- *The user interface shall be a responsive web application with a role-specific dashboard after login (donor, hospital/patient, staff, admin).*
- *Every screen shall show the application name, the logged-in user, a logout button and a navigation menu showing only the functions allowed for the user's role.*
- *Forms shall mark mandatory fields and shall show a clear error message next to any invalid field.*
- *Alerts (low stock, expiring units) shall be shown on the staff dashboard in a distinct colour with text, so that colour is not the only signal.*

Detailed screen layouts will be given in the design specification

Software Interfaces

- *Database: a relational database (PostgreSQL or MySQL) accessed from the application server. Data exchanged: users, donors, donations, blood units, blood requests and audit log entries.*
- *Email service (SMTP): the server sends password reset emails containing a one-time link.*
- *File export: the server generates CSV and PDF files for reports.*

Communications Interfaces

- *The browser and the server shall communicate over HTTPS (TLS 1.2 or higher).*
- *The user interface shall call the server through a REST API that exchanges JSON messages.*
- *Password reset emails shall be sent to the address registered by the user.*

Analysis Models

4.1 UML Use Case Diagram

The diagram shows the actors and use cases of the Blood Bank Management System.

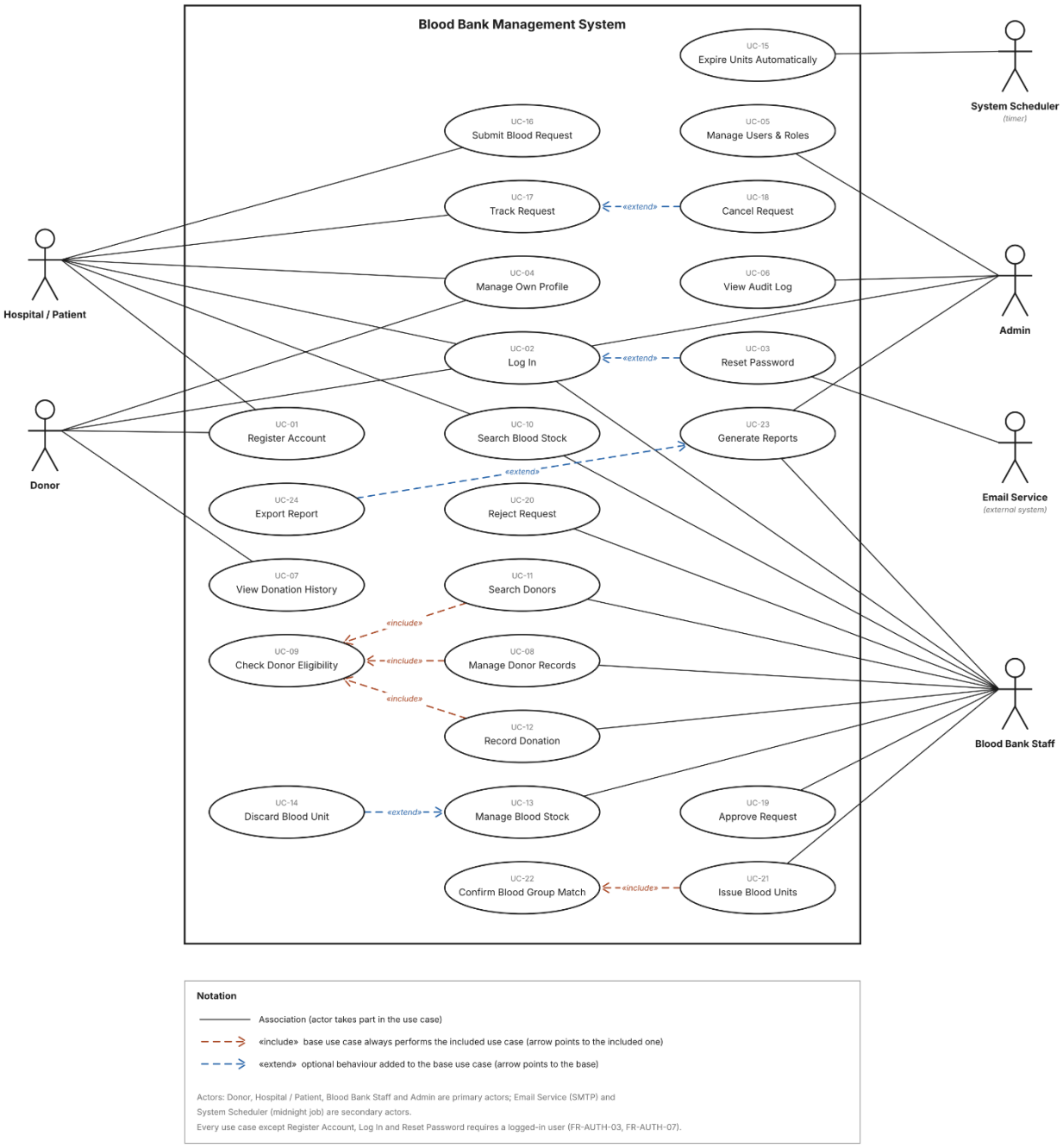


Figure 1: Use case diagram of the BBMS

4.2 Use Case Summary

ID	Use case	Actors	Description	SRS feature
UC-01	*Register / Login*	*Donor, Hospital / Patient, Blood*	*A user creates an account (donor, hospital/patient) or logs in with email and password.*	*5.1*

		Bank Staff, Admin		
UC-02	*Manage Users & Roles*	*Admin*	*Admin creates staff accounts, activates or deactivates accounts and assigns roles.*	*5.2*
UC-03	*Manage Own Profile*	*Donor, Hospital / Patient*	*User views and updates their own contact details.*	*5.1*
UC-04	*View Donation History*	*Donor*	*Donors view their own past donations.*	*5.3*
UC-05	*Search Blood / Donors*	*Hospital / Patient, Blood Bank Staff*	*User searches available blood stock (hospital, staff) or eligible donors (staff only) by blood group.*	*5.4*
UC-06	*Submit Blood Request*	*Hospital / Patient*	*Requester submits a request for blood units.*	*5.7*
UC-07	*Track / Cancel Request*	*Hospital / Patient*	*Requester checks the status of a request or cancels it.*	*5.7*
UC-08	*Process Blood Request*	*Blood Bank Staff*	*Staff approve, reject or issue a request.*	*5.7*
UC-09	*Manage Donor Records*	*Blood Bank Staff*	*Staff add, update and deactivate donor records and view eligibility.*	*5.3*
UC-10	*Record Donation*	*Blood Bank Staff*	*Staff record a donation and create a blood unit.*	*5.5*
UC-11	*Manage Blood Stock*	*Blood Bank Staff*	*Staff view stock, handle alerts and discard units.*	*5.6*
UC-12	*Generate Reports*	*Blood Bank Staff, Admin*	*User generates and exports stock, donation and request reports.*	*5.8*



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System Features

The functional requirements are grouped by system feature. Each requirement has a unique ID and is written to be testable.The owner of each feature is shown for team planning and will be removed in the final revision.

5.1 User Registration and Login

5.1.1 Description and Priority

Allows donors and hospital/patient users to create accounts and lets all users authenticate and maintain their own profile. Every other feature depends on it. Priority: High.

5.1.2 Stimulus/Response Sequences

1. *The user submits the registration form and the system validates the fields, creates the account and shows a confirmation.*
2. *The user enters email and password and the system authenticates and opens the dashboard for the user's role.*
3. *The user enters a wrong password repeatedly and the system locks the account temporarily and shows a lock message.*
4. *The user requests a password reset and the system sends a one-time reset link to the registered email.*

5.1.3 Functional Requirements

ID	***Requirement***
FR-AUTH-01	*The system shall allow a donor or hospital/patient user to register by providing name, email, phone number and password, and shall validate all mandatory fields before creating the account.*
FR-AUTH-02	*The system shall reject a registration whose email or phone number already exists and shall display the message "Account already exists".*
FR-AUTH-03	*The system shall authenticate a user by email and password and shall open the dashboard for the user's role within 3 seconds of a successful login.*

FR-AUTH-04	*The system shall lock an account for 15 minutes after 5 consecutive failed login attempts and shall display the remaining lock time.*

FR-AUTH-05	*The system shall let a user reset a forgotten password through a one-time link that expires after 15 minutes and can be used only once.*
FR-AUTH-06	*The system shall allow a logged-in donor or hospital/patient user to view and update their name, phone number and address.*
FR-AUTH-07	*The system shall end the session and require login again when the user logs out or after 15 minutes of inactivity.*

5.2 Admin Management

5.2.1 Description and Priority

Lets the administrator control who can use the system and in what role, and review what changed. Priority: Medium.

5.2.2 Stimulus/Response Sequences

1. *Admin opens the user list and the system shows all accounts with role and status, with search and filters.*
2. *Admin creates a staff account and the system validates the details and creates the account.*
3. *Admin deactivates or changes the role of an account and the system applies the change immediately and records it in the audit log.*

5.2.3 Functional Requirements

ID	***Requirement***
FR-ADM-01	*The admin shall be able to create a blood bank staff account by entering name, email, phone number and a temporary password.*

FR-ADM-02	*The system shall show the admin a searchable list of all user accounts that can be filtered by role and by status (active/inactive).*
FR-ADM-03	*The admin shall be able to activate or deactivate any user account, and the system shall prevent a deactivated user from logging in.*

FR-ADM-04	*The admin shall be able to assign or change a user's role to Donor, Hospital/Patient, Staff or Admin.*
FR-ADM-05	*The system shall record the user, action and timestamp of every admin action in an audit log that only the admin can view.*
FR-ADM-06	*The system shall prevent the deactivation or deletion of the last remaining active admin account.*

5.3 Donor Management

5.3.1 Description and Priority

Allows Blood Bank Staff to add, update and deactivate donor records and to see at any time whether a donor is eligible to donate. It also lets a donor view their own donation history. Donation recording (5.5) and donor search (5.4) depend on this feature. Priority: High.

5.3.2 Stimulus/Response Sequences

1. *Staff enter the donor details on the Add Donor form and submit; the system validates the fields, creates the record with a unique Donor ID and status Active, and shows a confirmation.*
2. *Staff open a donor record; the system shows the donor's details, donation history, eligibility status and next eligible date.*
3. *Staff edit a donor's details and save; the system validates the changes, saves them and keeps the donation history unchanged.*
4. *Staff deactivate a donor; the system marks the record Inactive and keeps all its data.*
5. *A donor logs in and opens Donation History; the system lists only that donor's own donations.*

5.3.3 Functional Requirements

ID	***Requirement***
FR-DON-01	*The system shall allow Blood Bank Staff to add a donor record by entering the mandatory fields listed in Appendix B.2 (full name, date of birth, gender, blood group, weight, phone number, city and PIN code), and shall generate a unique Donor ID and set the status to Active when the record is saved.*

FR-DON-02	*The system shall validate donor data before saving: the blood group shall be one of the eight valid blood groups; the date of birth shall not be in the future; the weight shall be a number between 30 and 250 kg; the phone number shall be exactly 10 digits; and the PIN code shall be exactly 6 digits. Each invalid field shall show an error message next to it, and a donor whose phone number already exists shall be rejected with the message "Donor already exists".*
FR-DON-03	*The system shall allow Blood Bank Staff to update any donor detail except the Donor ID. The update shall not change or delete the donor's existing donation records, and the system shall record the user and timestamp of the update.*
FR-DON-04	*The system shall determine donor eligibility whenever a donor record is displayed or a donation is recorded. A donor is Eligible only if the donor is Active, is aged 18 to 65 years inclusive, weighs at least 45 kg, and the gap since the last donation date is at least 90 days for a male donor or 120 days for a female or other donor. The system shall display "Eligible" or "Not eligible" with the next eligible date. These limits shall be configurable and not fixed in the code.*
FR-DON-05	*When a donor is Not eligible, the system shall display every failed criterion as a reason (for example, "Weight below 45 kg" or "Last donation was 40 days ago; next eligible date 15-11-2026") on the donor record.*

FR-DON-06	*The system shall allow a logged-in donor to view only their own donation history, showing Donation ID, collection date, blood group and volume for each donation, sorted with the newest first. If the donor has no donations, the system shall display "No donations recorded yet".*
FR-DON-07	*The system shall allow Blood Bank Staff to deactivate and reactivate a donor record, and shall not permit any user to delete a donor record. A deactivated donor shall not be recorded for new donations or appear in donor search results, but the donor's donation history shall be retained.*

FR-DON-08	*The system shall link a donor record to the donor's registered user account by matching the phone number or email address, so that a logged-in donor sees only the donation history of their own linked record. If no matching donor record exists, the donor's history page shall display "No donations recorded yet".*

5.4 Search Blood and Donors

5.4.1 Description and Priority

Lets Hospital/Patient users and Blood Bank Staff find available blood by blood group, and lets Blood Bank Staff find eligible donors by blood group and location. Hospitals use it to check availability before submitting a request (5.7). Priority: High.

5.4.2 Stimulus/Response Sequences

1. *A Hospital/Patient user or staff member selects a blood group and submits the stock search; the system shows the available units for that group with the earliest expiry date.*
2. *Staff enter a blood group and optionally a city or PIN code in donor search; the system lists the active, eligible donors that match.*
3. *The search matches nothing; the system shows a "No matching results found" message.*
4. *The search matches more than 20 records; the system shows the first page and controls to move between pages.*

5. *A Donor or Hospital/Patient user tries to search donors; the system refuses the request with an access denied message.*

5.4.3 Functional Requirements

ID	***Requirement***
FR-SRCH-01	*The system shall allow a Hospital/Patient user or Blood Bank Staff member to search blood stock by selecting one of the eight blood groups, and shall return only units whose status is Available and whose expiry date is later than the current date.*

FR-SRCH-02	*For each stock search, the system shall display the blood group, the number of available units and the earliest expiry date. For Blood Bank Staff the results shall also list each unit's Unit ID and expiry date. No donor identity shall be shown in stock results.*
FR-SRCH-03	*The system shall allow Blood Bank Staff to search donors by blood group and, optionally, by city or PIN code, and shall return only donors who are Active and currently Eligible under FR-DON-04. Each result shall show Donor ID, name, blood group, city, phone number and last donation date.*
FR-SRCH-04	*The system shall not allow Donor or Hospital/Patient users to search donors or view donor personal details. Any such request shall be refused with an "access denied" (HTTP 403) response and logged.*
FR-SRCH-05	*When a search returns no results, the system shall display the message "No matching results found" together with the criteria searched, and shall not show an error.*
FR-SRCH-06	*The system shall display search results in pages of 20 records, showing the total number of results and controls to move to the next and previous page.*

5.5 Blood Donation Recording

5.5.1 Description and Priority

Allows Blood Bank Staff to record a new blood donation for an eligible donor, perform pre-donation health screening validation, automatically update the donor's last donation date and eligibility, and generate a new blood unit record (with unique Unit ID, volume, blood group, collection date, and calculated expiry date). Priority: High.

5.5.2 Stimulus/Response Sequences

1. Staff select an eligible donor, enter donation details (collection date, volume in ml, blood group, screening result), and click Submit; the system validates the input, creates a donation record and a corresponding blood unit with status 'Available', updates the donor's last donation date, and displays a success message with the generated Donation ID and Unit ID.

2. Staff attempt to record a donation for a donor who is currently 'Not Eligible' under FR-DON-04; the system blocks submission and displays an error message stating the ineligibility reason (e.g., "Minimum gap between donations not met").

3. Staff record a donation where the donor fails pre-donation health screening; the system logs the screening failure, sets the donation status to 'Failed Screening', creates no blood unit, and updates the donor's record accordingly.

4. Staff select a recorded donation and submit an entry correction within 24 hours; the system validates the change, updates the record, logs the edit with staff user ID and timestamp, and updates stock records if volume/blood group was modified. 5.5.3 Functional Requirements

ID	***Requirement***
FR-DNR-01	*The system shall allow Blood Bank Staff to record a new donation by entering Donor ID, collection date, volume in ml (350ml or 450ml), blood group, and screening status (Passed / Failed).*
FR-DNR-02	*The system shall verify donor eligibility under FR-DON-04 before permitting a donation record to be saved; if the donor is marked "Not Eligible", the system shall prevent form submission and display the exact reason for ineligibility.*

FR-DNR-03	*Upon saving a successful donation (Screening Status = Passed), the system shall automatically generate a unique Blood Unit ID (formatted as UNT-XXXXX), set its status to Available, calculate its expiry date (35 days from collection date), and add it to the blood stock inventory.*
FR-DNR-04	*If pre-donation health screening fails (Screening Status = Failed), the system shall save the record with status Failed Screening, prevent creation of any blood unit, and log the reason in the donor's medical notes.*
FR-DNR-05	*Upon recording a successful donation, the system shall automatically update the donor's Last Donation Date field to the current collection date and recalculate the donor's next eligible donation date based on gender rules (FR-DON-04).*
FR-DNR-06	*The system shall allow Blood Bank Staff to edit a donation entry within 24 hours of creation for error correction; any modification shall be logged in the audit trail with staff ID, timestamp, original value, and modified value.*

5.6 Blood Stock Management

5.6.1 Description and Priority

Allows Blood Bank Staff to track and manage blood inventory across all eight blood groups, monitor unit statuses (Available, Reserved, Issued, Expired, Discarded), track shelf life, receive automated low-stock and near-expiry alerts, and discard unusable or expired units. Priority: High.

5.6.2 Stimulus/Response Sequences

1. Staff open the Blood Stock Management dashboard; the system displays total available units categorized by blood group (A+, A-, B+, B-, AB+, AB-, O+, O-), indicating status counts and highlighting badges for low stock.

2. *Staff select a specific blood group to inspect individual units; the system lists all corresponding units with Unit ID, collection date, expiry date, status, and location/shelf ID.*

3. *The system reaches midnight (00:00:00); an automated background process scans all blood units with status 'Available' or 'Reserved' whose expiry date is less than or equal to the current date, updates their status to 'Expired', and generates an automated notification.*

4. *Available stock for a blood group falls below the configured threshold (default: 5 units); the system generates a high-priority "Low Stock Alert" on the staff dashboard and highlights the affected blood group.*

5. *A blood unit comes within 7 days of its expiry date; the system tags the unit with an "Expiring Soon" warning badge on the stock view.*

6. *Staff select an expired, contaminated, or damaged unit and submit a discard request with a reason; the system updates the unit status to 'Discarded', records the discard reason and staff ID, and decrements available stock.*

5.6.3 Functional Requirements

ID	***Requirement***
FR-STK-01	*The system shall display blood stock inventory grouped by blood group (A+, A-, B+, B-, AB+, AB-, O+, O-) with counts for Available, Reserved, Issued, Expired, and Discarded units.*

FR-STK-02	*The system shall track the shelf life of each blood unit based on its collection date, setting the default expiration to 35 days from collection.*
FR-STK-03	*The system shall run an automated daily process at midnight to update all blood units whose expiry date has passed from status Available or Reserved to Expired.*
FR-STK-04	*The system shall trigger a visual "Low Stock Alert" on the staff dashboard whenever the count of Available units for any blood group drops below a configurable minimum threshold (default: 5 units).*

FR-STK-05	*The system shall display an "Expiring Soon" alert for any blood unit whose expiry date is within 7 days of the current date, prioritizing these units for issue or request fulfillment (First-Expiry, First-Out rule).*
FR-STK-06	*The system shall allow Blood Bank Staff to manually mark a unit as Discarded by selecting a mandatory reason (Expired, Hemolyzed, Contaminated, Leakage, Test Positive); the system shall update the unit status and log the action in the audit trail.*
FR-STK-07	*The system shall enforce non-negative stock counts and prevent any stock adjustment or unit allocation that would result in negative available inventory.*

5.7 Blood Request Handling

5.7.1 Description and Priority

Allows Hospital / Patient users to submit, track, and cancel blood requests, and enables Blood Bank Staff to process (review, approve, reject, and issue) blood requests using First-Expiry, First-Out (FEFO) rules and priority handling. Priority: High.

5.7.2 Stimulus/Response Sequences

1. Requester submits a blood request; the system validates the details, generates a unique Request ID, sets status to Pending, and adds it to the staff processing queue.

2. Requester views submitted requests; the system displays real-time status. If still Pending, the requester can cancel the request, updating status to Cancelled.

3. Staff review a Pending request and click Approve; the system checks available stock, reserves the matching units with the earliest expiry dates, and updates request status to Approved and unit statuses to Reserved.

4. Staff review a request and click Reject with a reason; the system sets status to Rejected, logs the reason, and displays it to the requester.

5. Staff issue blood for an Approved request; the system verifies blood group match, updates unit statuses to Issued, sets request status to Issued, and decrements stock.

5.7.3 Functional Requirements

ID	***Requirement***
FR-REQ-01	*The system shall allow a registered Hospital / Patient user to submit a blood request by providing patient name, blood group, units requested (1 to 10), required-by date, hospital name, contact number, and priority (Normal or Urgent).*
FR-REQ-02	*The system shall validate that required-by date is not in the past, units requested is between 1 and 10, blood group is valid, and contact number is exactly 10 digits before saving the request.*
FR-REQ-03	*Upon valid submission, the system shall generate a unique Request ID, set status to Pending, record requester ID and timestamp, and display the request in the staff pending queue.*
FR-REQ-04	*The system shall order pending requests for staff review by Priority (Urgent before Normal) and secondarily by submission timestamp in ascending order.*
FR-REQ-05	*The system shall allow Blood Bank Staff to approve a Pending request if sufficient Available units exist for the blood group, allocating available units using FEFO rules (earliest expiry first), setting unit status to Reserved, and updating request status to Approved.*
FR-REQ-06	*The system shall allow Blood Bank Staff to reject a Pending request by entering a mandatory rejection reason, setting request status to Rejected, and recording staff ID and timestamp.*
FR-REQ-07	*The system shall allow Blood Bank Staff to mark an Approved request as Issued upon physical handover, verifying blood group match and changing reserved units to Issued status.*

FR-REQ-08	*The system shall allow a Hospital / Patient user to track the status of their own requests and cancel any request that is currently in Pending status.*

5.8 Reports

5.8.1 Description and Priority

Allows Blood Bank Staff and Admins to generate, view, filter, and export reports for blood stock inventory, donation logs, and request history. Priority: Medium.

5.8.2 Stimulus/Response Sequences

1. The user selects a report type (Stock, Donation, or Request) and applies filters (date range, blood group).

2. The system fetches records, aggregates totals, and displays the report on screen within 10 seconds.

3. User selects Export as CSV or PDF; the system generates and downloads the formatted report file.

5.8.3 Functional Requirements

ID	***Requirement***
FR-RPT-01	*The system shall allow Blood Bank Staff and Admins to generate a Stock Report showing total available, reserved, issued, expired, and discarded units per blood group and earliest expiry date.*
FR-RPT-02	*The system shall allow Blood Bank Staff and Admins to generate a Donation Report listing donations within a date range, including Donation ID, Donor ID, blood group, volume, collection date, screening result, and total collection counts.*
FR-RPT-03	*The system shall allow Blood Bank Staff and Admins to generate a Request Report listing requests within a date range, including Request ID, hospital name, blood group, units, priority, status, request date, and decision date.*
FR-RPT-04	*The system shall allow users to filter any report by start date, end date, blood group, and status.*

FR-RPT-05	*The system shall allow users to export generated reports into CSV and PDF file formats.*
FR-RPT-06	*The system shall restrict report generation to Staff and Admin roles, returning access denied (HTTP 403) to unauthorized roles, and showing "No records found" when a search returns no data.*

Other Nonfunctional Requirements

Performance Requirements

ID	Requirement
NFR-PERF-01	*The system shall return search results within 3 seconds for at least 95% of searches with up to 10,000 donor records and 50,000 blood unit records. Rationale: staff may search while a patient is waiting.*
NFR-PERF-02	*The system shall load each page within 3 seconds for at least 95% of page loads on a 5 Mbps connection.*
NFR-PERF-03	*The system shall support 50 concurrent logged-in users with an average response time of no more than 3 seconds.*
NFR-PERF-04	*The system shall generate any report covering up to 12 months of data within 10 seconds.*

Safety Requirements

Blood is a medical product, so errors can harm patients. These requirements prevent unsafe use of the system.

ID	Requirement
NFR-SAF-01	*The system shall never issue a blood unit that is Expired, Discarded or reserved for another request; an attempt to do so shall be blocked with an error message.*



NFR-SAF-02	*The system shall require staff to confirm that the blood group of the unit matches the blood group of the request before the request can be marked Issued.*

NFR-SAF-03	*The system shall back up its database automatically every 24 hours and keep each backup for 30 days, so that at most one day of data is lost after a failure.*

Security Requirements

Security Objectives

ID	Requirement
SO-1	*Donor and patient personal and medical information shall be accessible only to the roles that need it.*
SO-2	*Donation, stock and request records shall not be changed by unauthorised users, and all permitted changes shall be traceable.*
SO-3	*Authorised users shall be able to use the system when they need it, and repeated failed logins or abusive requests shall not lock out other users.*
SO-4	*Every sensitive action shall be attributable to one identified user.*

Security Requirements

Each requirement shows in brackets the security objective it supports.

ID	Requirement
NFR-SEC-01	*The system shall require passwords of at least 8 characters containing an upper-case letter, a lower-case letter and a digit, and shall store them only as salted hashes (for example bcrypt), never in plain text. (SO-1)*
NFR-SEC-02	*The system shall enforce the user's role on the server for every page and API request; an unauthorised request shall return an "access denied" (HTTP 403) response and be logged. (SO-1, SO-2)*
NFR-SEC-03	*All communication between the browser and the server shall use HTTPS with TLS 1.2 or higher. (SO-1)*

NFR-SEC-04	*Sessions shall expire after 15 minutes of inactivity, and session cookies or tokens shall be marked HttpOnly and Secure. (SO-1, SO-3)*
NFR-SEC-05	*The system shall validate and sanitise all user input on the server and use parameterised database queries, so that SQL injection and cross-site scripting attempts are rejected. (SO-2)*
NFR-SEC-06	*The system shall log logins (successful and failed), role changes, donation entries, stock changes and request decisions with user ID and timestamp, and keep these logs for 1 year. (SO-4)*

NFR-SEC-07	*The system shall not write passwords, phone numbers, email addresses or addresses to application logs or show them in error messages. (SO-1)*

Software Quality Attributes

Relative preference: correctness and data safety come first, then usability, then performance.

ID	***Requirement***
NFR-QUAL-01	*A first-time staff user shall be able to register a donor and record a donation within 5 minutes using only on-screen guidance, verified by a usability test with at least 3 users.*
NFR-QUAL-02	*The system shall be available at least 99% of the time in any calendar month, excluding planned maintenance announced 24 hours ahead.*
NFR-QUAL-03	*Each donation, stock change and request decision shall be saved as one atomic transaction so that a failure never leaves stock and records out of step.*
NFR-QUAL-04	*The code shall be organised into separate modules per feature, and automated unit tests shall cover at least 70% of the business logic.*
NFR-QUAL-05	*The user interface shall work on the latest two versions of Chrome, Firefox and Edge and shall be usable on screens 360 pixels wide and above.*

Business Rules

ID	***Rule***
BR-01	*Only Staff can add donor records and record donations; only Admin can create staff accounts and change roles.*
BR-02	*A hospital/patient user can see only their own requests; a donor can see only their own donation history.*
BR-03	*A donor who is not eligible cannot donate until the next eligible date.*
BR-04	*Blood is issued from the unit with the earliest expiry date first.*
BR-05	*Urgent requests are processed before Normal requests.*
BR-06	*A request can be approved or rejected only by Staff, never by the requester.*

Domain requirements

- *The eight blood groups are A+, A-, B+, B-, AB+, AB-, O+ and O-.*
- *A blood unit has a limited shelf life and must not be issued after its expiry date.*
- *Donor eligibility is decided by age, weight and the time since the last donation.*

Other Requirements

Database Requirements

The system shall store its data in a relational database with at least these tables: User, Donor, Donation, BloodUnit, BloodRequest and AuditLog. Blood unit and request records shall reference the donation and the requester through foreign keys.

Legal Requirements

The system handles personal and health-related data and shall follow applicable Indian law, including the Digital Personal Data Protection Act, 2023 (exact obligations to be confirmed by the team).

Future Enhancements

- *Email or SMS notifications to donors and requesters.*
- *Donation appointment booking and donation camp management.*
- *Automatic blood group compatibility substitution.*
- *Mobile application.*

Appendix A: Glossary

Term	***Meaning***
BBMS	*Blood Bank Management System*

ABO / Rh	*The blood group system: types A, B, AB and O, each Rh positive (+) or negative (-).*
Blood group	*One of A+, A-, B+, B-, AB+, AB-, O+, O-.*
Blood unit	*One collected bag of blood tracked by a unique unit ID.*
Donor	*A person who donates blood.*
Eligibility	*Whether a donor currently meets the age, weight and donation-gap rules.*
Screening	*Tests on a donation that decide if it is safe to use.*
Expiry date	*The last date on which a blood unit may be issued.*
Issued	*Status of a unit handed over for a request.*

FEFO	*First Expiry, First Out: the unit expiring earliest is issued first.*
RBAC	*Role-based access control.*
SRS	*Software Requirements Specification.*
REST / JSON	*A web API style and the data format its messages use.*
HTTPS / TLS	*Encrypted web communication.*
CSV	*Comma-separated values, a simple spreadsheet file format.*
SRN	*Student registration number.*

Appendix B: Field Layouts

Field layouts and properties for the main records, followed by the fields required in each report.

B.1 User

Field	Length	Data Type	Description	Is Mandatory
User ID	10	Alphanumeric	Unique ID generated by the system	Y
Name	60	String	Full name	Y
Email	60	String	Login ID; must be unique	Y

Phone	10	Numeric	Mobile number; must be unique	Y
Password hash	60	String	Salted hash, never the password	Y
Role	15	String	Donor, Hospital/Patient, Staff or Admin	Y

Status	10	String	Active or Inactive	Y

B.2 Donor

Field	Length	Data Type	Description	Is Mandatory
Donor ID	10	Alphanumeric	Unique ID generated by the system	Y
Full Name	60	String	Donor name	Y
Date of Birth	8	Date	Used to calculate age	Y
Gender	10	String	Male, Female or Other	Y
Blood Group	3	String	One of the eight blood groups	Y
Weight (kg)	5	Numeric	Weight in kilograms	Y
Phone	10	Numeric	Contact number	Y
Email	60	String	Contact email	N
Address	120	String	Street address	N
City	40	String	City	Y
PIN Code	6	Numeric	Postal code	Y
Last Donation Date	8	Date	Updated on each donation	N
Status	10	String	Active or Inactive	Y

B.3 DONATION

Field	Length	Data Type	Description	Is Mandatory
Donation ID	12	Alphanumeric	Unique ID generated by the system	Y
Donor ID	10	Alphanumeric	Donor who donated	Y



Collection Date	8	Date	Date of donation	Y
Volume (ml)	4	Numeric	350 or 450	Y

Blood Group	3	String	Taken from the donor record	Y
Screening Status	10	String	Passed or Failed	Y
Donation Status	20	String	Recorded or Failed Screening	Y
Recorded By	10	Alphanumeric	User ID of the staff member	Y
Created At	14	Date-time	Used for the 24-hour correction rule	Y
Medical Notes	200	String	Reason, if screening failed	N

B.4 Blood Unit (from a donation)

Field	Length	Data Type	Description	Is Mandatory
Unit ID	12	Alphanumeric	Unique ID of the blood unit	Y
Donation ID	12	Alphanumeric	Donation this unit came from	Y
Blood Group	3	String	Blood group of the unit	Y
Volume (ml)	4	Numeric	Collected volume	Y
Collection Date	8	Date	Date of donation	Y
Expiry Date	8	Date	Collection date plus shelf life	Y
Status	10	String	Available, Reserved, Issued, Expired or Discarded	Y
Discard Reason	100	String	Reason if discarded	N

B.5 Blood Request

Field	Length	Data Type	Description	Is Mandatory
Request ID	12	Alphanumeric	Unique ID generated by the system	Y
Requester ID	10	Alphanumeric	User who submitted the request	Y
Patient Name	60	String	Patient needing blood	Y
Blood Group	3	String	Requested blood group	Y
Units	2	Numeric	Units requested (1 to 10)	Y
Required-by Date	8	Date	Date blood is needed	Y
Hospital Name	80	String	Hospital or place of care	Y

Contact Number	10	Numeric	Contact for this request	Y
Priority	10	String	Normal or Urgent	Y
Status	12	String	Pending, Approved, Rejected, Issued or Cancelled	Y
Rejection Reason	100	String	Reason if rejected	N

B.6 Report Requirements

Stock Report	Donation Report	Request Report
Blood Group	Donation ID	Request ID
Available units	Donor ID	Hospital Name
Reserved units	Blood Group	Blood Group
Issued units	Volume (ml)	Units requested
Expired / Discarded units	Collection Date	Priority

Earliest expiry date	Screening result	Status
Report date	Count by blood group	Request and decision dates

Appendix C: Requirement Traceability Matrix

The requirement IDs and descriptions are filled in. The architecture, design, code and test columns will be filled in as the architecture and design specification, the code and the test plan are written.

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-AUTH-01	Register account	TBD	TBD	TBD	TBD	TBD
2	FR-AUTH-02	Reject duplicate account	TBD	TBD	TBD	TBD	TBD
3	FR-AUTH-03	Login by role	TBD	TBD	TBD	TBD	TBD
4	FR-AUTH-04	Account lockout	TBD	TBD	TBD	TBD	TBD
5	FR-AUTH-05	Password reset	TBD	TBD	TBD	TBD	TBD
6	FR-AUTH-06	Manage own profile	TBD	TBD	TBD	TBD	TBD

7	FR-AUTH-07	Logout and session end	TBD	TBD	TBD	TBD	TBD
8	FR-ADM-01	Create staff account	TBD	TBD	TBD	TBD	TBD
9	FR-ADM-02	View and filter users	TBD	TBD	TBD	TBD	TBD
10	FR-ADM-03	Activate / deactivate user	TBD	TBD	TBD	TBD	TBD
11	FR-ADM-04	Assign role	TBD	TBD	TBD	TBD	TBD
12	FR-ADM-05	Audit log	TBD	TBD	TBD	TBD	TBD
13	FR-ADM-06	Protect last admin	TBD	TBD	TBD	TBD	TBD
14	FR-DON-01	Add donor record	TBD	TBD	TBD	TBD	TBD
15	FR-DON-02	Validate donor data	TBD	TBD	TBD	TBD	TBD
16	FR-DON-03	Update donor and history	TBD	TBD	TBD	TBD	TBD

17	FR-DON-04	Eligibility check	TBD	TBD	TBD	TBD	TBD
18	FR-DON-05	Show ineligibility reason	TBD	TBD	TBD	TBD	TBD
19	FR-DON-06	Donor views own history	TBD	TBD	TBD	TBD	TBD
20	FR-DON-07	Deactivate, not delete	TBD	TBD	TBD	TBD	TBD
21	FR-SRCH-01	Search stock by group	TBD	TBD	TBD	TBD	TBD
22	FR-SRCH-02	Stock result details	TBD	TBD	TBD	TBD	TBD
23	FR-SRCH-03	Search donors	TBD	TBD	TBD	TBD	TBD
24	FR-SRCH-04	Hide donor details	TBD	TBD	TBD	TBD	TBD
25	FR-SRCH-05	No result message	TBD	TBD	TBD	TBD	TBD
26	FR-SRCH-06	Paginate results	TBD	TBD	TBD	TBD	TBD
27	FR-DNR-01	Record donation	TBD	TBD	TBD	TBD	TBD
28	FR-DNR-02	Block ineligible donor	TBD	TBD	TBD	TBD	TBD
29	FR-DNR-03	Create blood unit	TBD	TBD	TBD	TBD	TBD
30	FR-DNR-04	Failed screening	TBD	TBD	TBD	TBD	TBD
31	FR-DNR-05	Update last donation	TBD	TBD	TBD	TBD	TBD
32	FR-DNR-06	Correct entry	TBD	TBD	TBD	TBD	TBD
33	FR-STK-01	Stock by group and status	TBD	TBD	TBD	TBD	TBD
34	FR-STK-02	Expiry date	TBD	TBD	TBD	TBD	TBD
35	FR-STK-03	Auto-expire units	TBD	TBD	TBD	TBD	TBD
36	FR-STK-04	Low-stock alert	TBD	TBD	TBD	TBD	TBD
37	FR-STK-05	Expiry alert	TBD	TBD	TBD	TBD	TBD

38	FR-STK-06	Discard unit	TBD	TBD	TBD	TBD	TBD
39	FR-STK-07	No negative stock	TBD	TBD	TBD	TBD	TBD



40	FR-REQ-01	Submit request	TBD	TBD	TBD	TBD	TBD
41	FR-REQ-02	Validate request	TBD	TBD	TBD	TBD	TBD
42	FR-REQ-03	Assign ID, set Pending	TBD	TBD	TBD	TBD	TBD
43	FR-REQ-04	Pending queue order	TBD	TBD	TBD	TBD	TBD
44	FR-REQ-05	Approve and reserve	TBD	TBD	TBD	TBD	TBD
45	FR-REQ-06	Reject with reason	TBD	TBD	TBD	TBD	TBD
46	FR-REQ-07	Issue units	TBD	TBD	TBD	TBD	TBD
47	FR-REQ-08	Track and cancel	TBD	TBD	TBD	TBD	TBD
48	FR-RPT-01	Stock report	TBD	TBD	TBD	TBD	TBD
49	FR-RPT-02	Donation report	TBD	TBD	TBD	TBD	TBD
50	FR-RPT-03	Request report	TBD	TBD	TBD	TBD	TBD
51	FR-RPT-04	Filter reports	TBD	TBD	TBD	TBD	TBD
52	FR-RPT-05	Export reports	TBD	TBD	TBD	TBD	TBD
53	FR-RPT-06	Report access and empty	TBD	TBD	TBD	TBD	TBD
54	NFR-PERF-01	Search response time	TBD	TBD	TBD	TBD	TBD
55	NFR-PERF-02	Page load time	TBD	TBD	TBD	TBD	TBD
56	NFR-PERF-03	Concurrent users	TBD	TBD	TBD	TBD	TBD
57	NFR-PERF-04	Report generation time	TBD	TBD	TBD	TBD	TBD
58	NFR-SAF-01	No unsafe issue	TBD	TBD	TBD	TBD	TBD
59	NFR-SAF-02	Confirm group on issue	TBD	TBD	TBD	TBD	TBD
60	NFR-SAF-03	Daily backup	TBD	TBD	TBD	TBD	TBD
61	NFR-SEC-01	Password policy and hashing	TBD	TBD	TBD	TBD	TBD
62	NFR-SEC-02	Role-based access control	TBD	TBD	TBD	TBD	TBD
63	NFR-SEC-03	Encrypted transport	TBD	TBD	TBD	TBD	TBD

64	NFR-SEC-04	Session security	TBD	TBD	TBD	TBD	TBD
65	NFR-SEC-05	Input validation	TBD	TBD	TBD	TBD	TBD
66	NFR-SEC-06	Audit trail	TBD	TBD	TBD	TBD	TBD
67	NFR-SEC-07	No personal data in logs	TBD	TBD	TBD	TBD	TBD
68	NFR-QUAL-01	Usability	TBD	TBD	TBD	TBD	TBD

69	NFR-QUAL-02	Availability	TBD	TBD	TBD	TBD	TBD
70	NFR-QUAL-03	Data consistency	TBD	TBD	TBD	TBD	TBD
71	NFR-QUAL-04	Maintainability and testability	TBD	TBD	TBD	TBD	TBD
72	NFR-QUAL-05	Browser compatibility	TBD	TBD	TBD	TBD	TBD